

RAFFLES INSTITUTION
2018 Year 6 Preliminary Examination Suggested Answers

Higher 1

CANDIDATE
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BIOLOGY

8876/02

Paper 2 Structured and Free-response Questions

13th September 2018

2 hours

Candidates answer on the Question Paper.
Additional Materials: Writing paper.

READ THESE INSTRUCTIONS FIRST

Write your index number, CT group & name in the spaces at the top of this page.

Write in dark blue or black pen.

You may use a HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

DO NOT WRITE IN ANY BARCODES

Section A

Answer **all** questions in the spaces provided on the Question Paper.6

Section B

Answer any **one** question in the writing paper provided.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units

At the end of the examination, **hand in your essay question SEPARATELY.**

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
SECTION A	
1	/12
2	/9
3	/11
4	/7
5	/6
SECTION B	
6 or 7	/15
Total	/60

This document consists of **10** printed pages.



Raffles Institution
Internal Examination

[Turn over

2018 Prelim H1 Answers:

1.	D	6.	D	11.	D	16.	B	21.	B	26.	
2.	A	7.	C	12.	C	17.	D	22.	A	27.	
3.	C	8.	D	13.	D	18.	C	23.	B	28.	
4.	B	9.	B	14.	C	19.	D	24.	C	29.	
5.	A	10.	B	15.	C	20.	A	25.	C	30.	

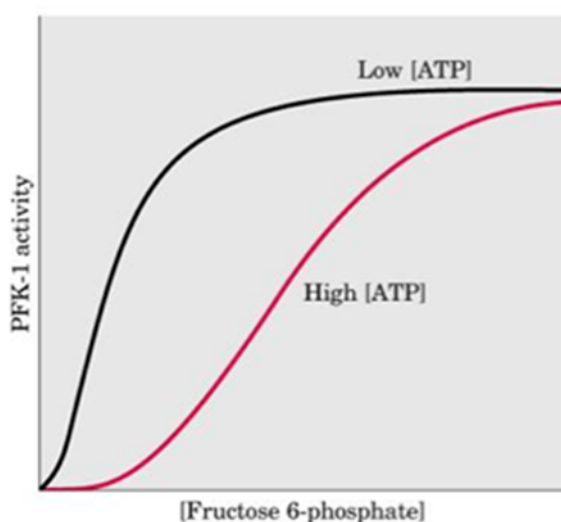
2018 Prelim H2 Suggested Answers

SECTION A

Answer **all** questions.

1

- (a) (i) On Fig 1.1, draw a graph to show the effect of high ATP on PFK activity [1]



- (ii) Explain the graph you have drawn. [3]

- 1) High levels of ATP inhibits PFK activity by lowering its affinity for fructose 6-phosphate.
- 2) ATP acts as an allosteric inhibitor which binds to an allosteric site which inhibits the PFK activity
- 3) $[V_{\max}]$: inhibition can be overcome by high substrate concentrations;
- 4) [idea of cooperativity] binding of substrate to 1 subunit changes conformation of other subunits such that it becomes easier for substrate to bind;

- (iii) Name 1 molecule that will act as allosteric activator of PFK activity.[1]

AMP/ADP

(b)

(i) Describe the structure of PFK. [4]

- 1) PFK has a quaternary* structure made up of 4 polypeptide chains
- 2) Each of which has primary* structure which is the number and sequence of amino acid
- 3) The secondary* structure of PFK is made up of alpha helix and beta pleated sheets (name at least 1)
- 4) The tertiary* structure of PFK is formed by interaction of R groups of amino acid namely hydrogen bond, hydrophobic interaction, disulfide bond and ionic bond within the same polypeptide
- 5) The overall conformation of PFK is maintained by its primary, secondary and tertiary structure in addition to the interaction among the 4 polypeptide chains

(ii) Explain how a change in pH affect the PFK activity.[3]

- 1) Affects ionisation of R groups* of aa.
- 2) A change in pH will disrupt the ionic and hydrogen bond* that maintain the 3D conformation of active site of PFK
- 3) which will result in a change in 3D conformation of active sites*, no longer complementary in shape and charge to substrate / substrate cannot fit, hence
- 4) less enzyme-substrate complex* formed hence reducing the PFK activity

[Total : 12]

2

(a)

Discuss the role of centromere in the production of gametes.[3]

1. They allow (proteins called kinetochore proteins, and subsequently) spindle fibres, to attach; (MUST HAVE)2a. resulting in proper alignment of chromosome singly in on equator in metaphase ;2b. so that sister chromatids / daughter chromosome can be equally separated to
opposite poles during Anaphase;3a. resulting in proper alignment of bivalents on equator in metaphase ;3b. so that homologous chromosomes can be equally separated to opposite poles during Anaphase 1;

(b)

(i) What is the mode of inheritance of the disease WAS? [1]

Sex-linked recessive

(ii) Use suitable symbols to represent the alleles of the gene controlling the disease WAS[1]

 X^W : Dominant allele that codes for no WAS on X chromosome X^w : recessive allele that codes for disease WAS on X chromosome;Must show both allelesMust have X chromosome on symbol

(iii) Use a genetic diagram to explain the results of couple 2. [4]

1] parental phenotype and genotype;

P phenotype: wavy-hair female WAS sufferer

P genotype:

 $X^wX^wH^SH^C$

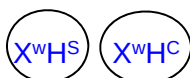
x

wavy-hair normal male ;

x

 $X^WYH^SH^C$

Gametes (n):



x

[1] all correct gametes (circled) in P and F₁ gen

		Male gamete			
		X^WH^S	YH^S	X^wH^C	YH^C
Female gametes	X^dH^S	$X^WX^wH^SH^S$ Normal Straight hair female	$X^wYH^SH^S$ WAS sufferer, straight hair male	$X^WX^dH^SH^C$ Normal wavy hair female	X^wYH^S WAS suf wavy hair

	X^dH^C	$X^W X^w H^S H^C$ Normal wavy hair female	$X^w Y H^S H^C$ WAS sufferer, wavy hair male	$X^W X^d H^C H^C$ Normal curly hair female	$X^w Y H^C H^C$ WAS sufferer, curly hair male
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[1] Punnett square with correct genotypes and related phenotypes (marker to note, don't need phenotype)

offspring genotypic ratio	$X^W X^d H^S H^S$	$X^W X^w H^S H^C$	$X^W X^w H^C H^C$	$X^w Y H^S H^S$	$X^w Y H^S H^C$	$X^w Y H^C H^C$
offspring phenotypic ratio	Normal Straight hair female	Normal wavy hair female	Normal curly hair female	WAS sufferer, straight hair male	WAS sufferer, wavy hair male	WAS suf hair ma
	1	2	1	1	2	1

[1] Correct F_2 genotypic and phenotypic ratio

[Total : 9]

3

- (a) (i) With reference to **Fig. 3.1**, state the best conditions for the growth of cucumber.[1]
0.13% carbon dioxide, 30°C, above 5 arbitrary units of light;
- (ii) With reference to **Fig. 3.1**, explain reasons for the difference between curves B and C.[3]
- Curve B reaches a plateau at rate of photosynthesis of $95 \text{ mm}^3\text{cm}^{-2}\text{h}^{-1}$ at lower light intensity of 3 units, whereas Curve C reaches a higher maximal rate of photosynthesis of $210 \text{ mm}^3\text{cm}^{-2}\text{h}^{-1}$ at higher light intensity of 5 units; (idea of : max rate being different, with values)
 - due to presence of higher CO_2 concentration at 0.13 (graph C);
 - OR
 - Curve B: reaches max rate at light intensity of 3 units, while Curve C reaches max rate at light intensity of 5 unit
 - light intensity is no longer a limiting factor as any further increase in light intensity does not affect rate of photosynthesis. Hence, CO_2 concentration becomes limiting;
- (max 2 from 1 and 2 or 3 and 4)
- Plant C achieve higher rates of photosynthesis as compared to B, because higher CO_2 / substrate concentration during carbon fixation* in Calvin cycle.
 - results in higher frequency of effective collision* with enzyme Rubisco (and RuBP), thus more enzyme-substrate complexes* being formed;
- (b) With reference to **Fig. 3.2**,
- (i) explain why most plants are characteristically green [2]

1. Percentage of green light absorbed by (all 3 pigments – marker discretion) is very low at less than 5%;
2. Green light is reflected / transmitted;

For reference: Blue ~ 490–450 nm, red ~ 635-700 nm

(ii) explain the shape of the action spectrum. [3]

1. Peak(s) of action spectrum/ highest rate of photosynthesis, correspond to absorption peak(s) (Idea of : action spectrum = sum of absorption of all pigments) ;
2. Light absorbed by different photosynthetic pigments is used for photoactivation/ photosynthesis
3. thus higher percentage of light absorption leads to higher rate of photosynthesis in red and blue light; (idea of : chl a peak corresponding to action spectrum peak)
4. carotenoids / chl b absorbed at blue and red light, broadening the range of wavelength that can be used for photosynthesis
5. Region between 500 to 600 nm (R: green) is not an exact match between absorption and action spectra due to role of other accessory pigments

(c) In conducting experiments to obtain the action spectrum for photosynthesis, explain why use of organic solvents such as acetone or ether is necessary to extract photosynthetic pigments. [2]

1. Chlorophyll pigments are membrane-bound (A: thylakoid and chloroplast membranes);
2. Only organic solvents can dissolve the membrane, that is composed of a phospholipid bilayer;

[Total: 11]

4

(a) Describe two changes to the bones in the feet of horses that have taken place over the past 50 million years.[2]

1. The bones became larger/longer/thicker;
2. The number of bones became fewer/idea of fewer toes or bones fused;
3. Fewer bones touched the ground/lower surface area in contact with the ground;

(b)

(i) Explain **one** advantage to *Eohippus* of the arrangement of bones in its feet.[1]

1. With a larger surface area (inference of data) in contact with the ground, it can distribute its weight and not sink into the soft mud/ can escape predators (advantage);
Must have context specific info and clear advantage.

(ii) Based on the information given, explain how the changes in the arrangement of the foot bones of horses support Darwin's theory of evolution by natural selection.[4]

1. It shows descent with modification from a common ancestor;
2. There was variation in the size/number/arrangement of foot bones in the population; (context specific)
3. The environment changed from swampy to hard and firm ground; (envt change,

- quote)
4. They faced selection pressure of predation; (idea of selection pressure, quote)
 5. Natural selection selected for individuals who had larger/fewer foot bones as they are; (selected traits)
 6. Able to run faster on firm ground to avoid predation; (suggest why trait is an advantage)
 7. Those who survive, are able to reproduce and pass on their alleles for larger/fewer bones to the next generation;

[Total: 7]

5

- (a) With reference to **Fig. 5.1**, suggest why the sea ice extent in March is different from that in September. [1]

March – maximum sea ice after sea water freezes in the winter/colder months

September – minimum sea ice after melting in the summer/hotter months

- (b) (i) Identify X shown in **Fig. 5.2**. [1]

X: duration of sea ice melt;

- (i) Describe how X changes from 1979 to 2015. [1]

1. X increased by about a month from 1979 to 2015
2. Sea ice melt season is starting earlier and ending later, melt season spanning from mid june to mid sept in 1979, to late may to late sept in 2015;

- (ii) Explain how this is linked to the changes in sea ice extent as shown in Fig. 5.1 and Fig. 5.3. [3]

1. Longer sea ice melt season results in more ice melting, lowering minimum sea ice extent observed in September; (Fig. 5.1)
2. This also means that less time is available for formation of sea ice in the winter months;
3. Resulting in a consistent decrease in maximum sea ice extent seen in March (Fig. 5.1);
4. Longer duration of sea ice melt resulted in less ice remaining frozen past the melt season;
5. As seen in the decrease in the amount / proportion of multi-year old ice (Fig. 5.3);
6. Which implies thinning of ice cover over the years;

Section C

Answer one question in this section.

Write your answers on the separate writing paper provided.

Your answers should be illustrated by large, clearly labeled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections (a), (b) etc., as indicated in the question.

6 (a) Account for the importance of hydrogen bonds in biological organisms. [7]

Ref to H2 answers

(b) Describe the main features of replication of DNA. [8]

1. Replication of DNA is semi-conservative* where both of the original strands of the double helix separate* and act as templates* for the synthesis of two new strands;
2. This gives rise to new/daughter DNA molecule which consists of one original and one newly synthesised strand;
3. replication of DNA begins at the origin of replication* where enzyme helicase* will bind and unzip* the DNA molecule by breaking hydrogen bonds* between complementary base pairs;
4. separated strands of DNA interact with the single stranded DNA binding proteins* so that it will remain single stranded and can serve as a template* for replication;
5. Topoisomerase* relieves "overwinding" strain ahead of replication forks by breaking, swivelling and rejoining DNA strands;
6. enzyme primase* catalyses the synthesis of a short RNA primer;
7. Complementary base pairing of free deoxyribonucleotide triphosphates (dNTP) such as dATP, dTTP, dGTP and dCTP with the nucleotides on the template occurs;
8. Purines Adenine and Guanine base pairs with pyrimidines Thymine and Cytosine respectively;
9. DNA polymerase* catalyses the formation of phosphodiester bond* linking DNA nucleotides;
10. and the new DNA strand is synthesised in the 5' to 3' direction*;
11. RNA primers* will then be removed and replaced with DNA by another DNA polymerase;
12. one of the daughter strands known as the leading strand* will be synthesised continuously;
13. the other strand known as the lagging strand* is synthesised discontinuously, giving rise to Okazaki fragments*;
14. DNA ligase* catalyses the formation of phosphodiester bond* between the Okazaki fragments;

R: If use ATGC without spelling the full word. Molecule of DNA refers to double stranded helix. Strand refers to a single polynucleotide. No exceptions accepted

OR

7 (a) Discuss the diverse roles of membranes in a cell. [7]

Ref to H2 answers

(b) Describe the major properties of enzymes and discuss their mode of action. [8]

Properties of enzymes

1. Most enzymes are globular proteins which are soluble and can be transported
2. Enzymes have active sites which are complementary in shape and charge to substrates
3. Enzymes can be denatured by temperatures and pH
4. Enzymes are required in small amount and can be reused

5. Enzymatic reactions can be inhibited by competitive or non-competitive inhibitors which slows down the rate of reaction of enzymes
6. Enzymatic reactions can be regulated by end-product inhibition

Mode of action of enzymes

7. Enzymes have an specific* active site* which is complementary in shape/conformation and charge* to the substrate;
8. Effective collisions between enzyme and substrate form a temporary enzyme-substrate complex*;
9. Based on the lock and key* hypothesis, enzyme is the lock* and substrate is the key* (lock and key hypothesis) Or
10. Based on the induced fit* hypothesis, substrate induces a change in shape in enzyme active site so that active site is a more precise fit for substrate for effective catalysis.
11. Enzyme-substrate complex held together by weak interactions e.g. hydrogen, ionic bonds, hydrophobic interactions*
12. Catalysis - Enzyme lowers the activation energy* barrier by
13. Aligning substrates next to each other (close proximity) in active site for reaction to occur (any one of following elaboration)
 - Strain on bonds to be broken / distorts the substrate and reduces activation energy to achieve transition state
 - Orientates substrate such that its bonds are exposed to attack
 - Provide a favorable microenvironment
 - R-groups of amino acid residues in active site participate in direct catalysis – e.g. Acid-base catalysis
14. Products no longer fit active site and are released → enzyme is unchanged and can be used again;